

JavaScript		
Variables (let, const, var)		
Data types and type coercion		
Functions and arrow functions		
Object literals, methods, this keyword		
Destructuring		
Error Handling, Custom errors		
template literals		
Closures and IIFE (Immediately Invoked Function Expressions)		
Objects & Prototypes		
Prototypal inheritance		
Object.create, Object.assign, Object.keys/values/entries		
Callbacks and callback hell		
Selecting elements (getElementById, querySelector)		
Event delegation		
shadowing		
high order function, pure function, first class function		
Promise chaining and error handling		
Arrays and objects manipulation and all methods		
Promises and async/await		
Event loop and concurrency model		
rest and spread operator		
web apis and web hooks		
LocalStorage / SessionStorage/cookies		
function declaration function expression		
map,filter,reduce method		
function composition		
call(),bind(),apply()		
shallow copy and deep copy		
getter and setter		

static method		
curring		
setTimeout and setInterval		
generator function		
event bubbling and event capturing		
ReactJs		
React fundamentals (components, JSX)		
Functional vs Class components		
Props and State		
Component lifecycle		
Conditional rendering		
Lists and keys		
Forms and controlled components		
Event handling		
Hooks (useState, useEffect, useRef, useMemo, useCallback, useContext, custom hooks)		
Context API		
Lifting state up		
Code splitting and lazy loading		
Error boundaries		
Prop drilling and context		
CSS in React (CSS Modules, Styled Components, Tailwind, Emotion)		
State management (Redux, Zustand, Jotai, Recoil)		
Redux Toolkit		
Async actions (Redux Thunk, Redux Saga)		
Data fetching (fetch, axios, react-query)		
Component composition and reusability		
Performance optimization (memoization, virtualization)		

NextJs

What is Next.js and why use it

File-based routing

Pages and components structure

Pre-rendering (SSG, SSR, ISR)

Data fetching methods (getStaticProps, getServerSideProps, getStaticPaths)

API routes in Next.js

Dynamic routes

Image optimization (next/image)

Link and navigation (next/link, useRouter)

Metadata and Head component (next/head)

Middleware in Next.js

Environment variables

Authentication (NextAuth.js)

State management in Next.js

Error handling and custom pages (404, 500)

TypeScript

What is TypeScript? Why use it?

How TypeScript compiles to JavaScript (the .ts → .js process)

Setting up a project (tsconfig.json)

Running TS with ts-node

Basic Types

Arrays & Tuples Typed arrays (string[], Array<number>)

Tuples ([string, number])

Readonly arrays (readonly string[])

Async programming with Node		
Error handling		
Middleware concept		
Node REPL		
Performance and optimization		
Routing (GET, POST, PUT, DELETE)		
Middleware (application-level, router-level, error-handling)		
Request and response objects		
Handling forms and JSON data		
URL parameters and query strings		
Express Router		
CORS (Cross-Origin Resource Sharing)		
Authentication (JWT, OAuth, cookies, sessions)		
Authentication & Authorization (JWT, sessions)		
Validation and sanitization (express-validator)		
File uploads (Multer)		
Security best practices (helmet, rate limiter)		
RESTful API design principles		
MVC architecture with Express		
Connecting Express with databases (MongoDB, PostgreSQL, MySQL)		
API versioning		
Caching (Redis)		
CI/CD basics (GitHub Actions, Jenkins)		
Redis		
Prisma		
NestJs		

How Nest works under the hood (modules, DI container)		
Modules		
Controllers		
Providers		
Services		
What is IoC (Inversion of Control)		
Dependency Injection (DI)		
useClass, useValue, useFactory, useExisting		
Injection scopes		
Default (singleton)		
Transient		
Request-scoped		
Pipes, Guards, Filters, Interceptors		
Pipes (ValidationPipe (class-validator), Transformation pipe, Custom pipes)		
Guards (AuthGuard, RoleGuard, Custom guard using CanActivate)		
Mongodb		
mongodb		
collection,documents		
CURD operations		
indexes(createIndex({email:1}))		
Aggregation		
relationships in mongodb(embedding, referencing)		
Replication		
sharding		

Postman		
postman		
Agile methodology		
websocket		
websocket		
ssh		
monorepo		
microservices		
email, zod , validation , api best practice , worker thread , clustering ,		